

MSc Dissertation Research Proposal: Efficient Small Language Models under Stacked Compression

TL;DR

- Build the dissertation around ONE defensible gap: how compression methods INTERACT when stacked (not studied in isolation) on sub-8B models, measured on accuracy AND calibration/robustness AND real hardware cost (FLOPs, tokens/sec, memory, energy). This is genuinely open, single-student-sized, and runs on a T4/L4/MPS.
- Four RQs form a chain: RQ1 baseline characterisation, RQ2 head-to-head single-method comparison, RQ3 interaction/ordering analysis (the novel core), RQ4 on-device deployment validation on Apple Silicon.
- Everything fits Colab Pro plus a MacBook Pro. Use the Qwen2.5 spine (0.5B/1.5B/7B, Apache 2.0) as the workhorse, lm-evaluation-harness for accuracy, and produce a reproducible benchmark artifact that doubles as a strong CV piece and a realistic workshop paper.

Title

"Stacked and Sequenced: An Empirical Study of Interacting Compression Methods for Sub-8B Language Models on Consumer Hardware"

Problem statement and research gap

Quantisation, pruning, knowledge distillation (KD), and parameter-efficient fine-tuning (PEFT such as LoRA/QLoRA/DoRA) are each well studied in isolation, mostly on large (30B-70B) models. Two things are genuinely open for a single student. First, compression dynamics for small language models (SLMs, roughly 0.5B-8B) are under-explored relative to LLMs, and small models behave differently: in "A Comprehensive Evaluation of Quantized Instruction-Tuned Large Language Models" (Jin et al., arXiv:2409.11055), 4-bit weight-only degradation stays within a negligible-to-moderate range (under 4%) for larger models, and the paper reports that instruction-following (IFEval) and hallucination/truthfulness (TruthfulQA) tasks suffer most under quantisation, which is exactly where small models are already weak. "Revisiting Pruning vs Quantization for Small Language Models" (Zhou, Kurz, Zhao, Findings of EMNLP 2025, 2025.findings-emnlp.645) states plainly that SLM "compression dynamics remain underexplored compared to Large Language Models." Second, the INTERACTION and ORDERING of stacked compression methods (prune-then-quantise, distil-then-quantise, LoRA on an already-quantised backbone) is not systematically characterised at sub-8B scale, and almost never with calibration or robustness as outcomes rather than accuracy alone. "A Systematic Study of Compression Ordering for Large Language Models" (Chhawri, Mahadik, Rooj, arXiv:2511.19495) confirms "the existing literature provides limited guidance on the optimal ordering of these techniques when applied sequentially on small scale LLMs," but demonstrates this on only a single 3B model. The

open gap: a controlled, reproducible, multi-model study of stacked compression at sub-8B scale, quantifying synergy versus antagonism across accuracy, calibration, robustness, and measured hardware cost on a single consumer GPU and Apple Silicon.

arxiv + 3

The four research questions

RQ1 (Baseline characterisation)

Question: For a fixed family of open-weight sub-8B models, what is the accuracy-efficiency-calibration profile at full precision (FP16/BF16) across reasoning, instruction-following, and domain tasks?

- **Hypothesis:** Larger models in the family dominate on accuracy but at proportionally worse latency/memory; calibration error is non-trivial even before compression.
- **Independent variables:** model, model size, task.
- **Dependent variables:** task accuracy, expected calibration error (ECE), FLOPs/token, tokens/sec, time-to-first-token (TTFT), peak memory.
- **Experiments:** run each base model through lm-evaluation-harness on the chosen benchmarks; profile FLOPs and latency/memory; compute calibration from logits.
- **"Good result" even if hypothesis wrong:** a clean, reproducible baseline table is itself a deliverable. A surprising finding (e.g. a smaller model matching a larger one on a task) is a positive contribution.

RQ2 (Single-method comparison)

Question: Applied individually, how do quantisation (GPTQ, AWQ, bitsandbytes NF4), pruning (Wanda, SparseGPT), and PEFT (LoRA/QLoRA/DoRA) trade accuracy against efficiency on these models?

- **Hypothesis:** 4-bit weight-only quantisation gives the best accuracy retention per unit memory saved; unstructured pruning gives little real speedup on consumer GPUs.
- **Independent variables:** compression method, bit-width/sparsity/rank, model.
- **Dependent variables:** the full RQ1 metric set plus model size on disk.
- **Experiments:** apply each method at matched compression ratios; re-evaluate. Expect AWQ to retain more accuracy than GPTQ at 4-bit weight-only: Jin et al. (arXiv:2409.11055) report "AWQ generally shows less accuracy degradation compared to GPTQ" and "outperformed GPTQ in terms of MT-Bench scores."
- **"Good result" even if hypothesis wrong:** a Pareto frontier plot per model. Negative results (e.g. pruning underperforming, or GPTQ matching AWQ on your models) are valuable and reportable.

RQ3 (Interaction and ordering: the novel core)

Question: When two compression methods are stacked, are effects additive, synergistic, or antagonistic, and does ORDER matter (prune-then-quantise vs quantise-then-prune, LoRA-on-quantised vs quantise-after-LoRA)?

- **Hypothesis:** Order matters; applying quantisation last (after pruning/distillation/LoRA) preserves more accuracy than quantising first (consistent with Chhawri et al. finding that early quantisation caused severe degradation via irreversible information loss).
- **Independent variables:** method pair, ordering, model.
- **Dependent variables:** accuracy, ECE, robustness under input perturbation, plus the efficiency set. Define an interaction effect as observed stacked accuracy minus predicted-from-individual accuracy.
- **Experiments:** factorial design over method pairs and orderings at fixed compression budget; compute interaction terms; repeat with multiple seeds where training is involved.
- **"Good result" even if hypothesis wrong:** a quantified interaction matrix is the contribution regardless of direction. Finding no interaction is itself a clean, publishable negative result.

RQ4 (Generalisation / on-device deployment validation)

Question: Do the best stacked configurations from RQ3 transfer to Apple Silicon (MPS/MLX/GGUF via llama.cpp), and how do measured on-device latency, memory, and energy compare to the CUDA estimates?

- **Hypothesis:** Accuracy rankings transfer, but efficiency rankings shift because Apple Silicon is memory-bandwidth bound and unstructured sparsity yields no speedup without special kernels.
- **Independent variables:** deployment runtime (CUDA transformers/vLLM vs MLX vs llama.cpp Metal), configuration.
- **Dependent variables:** tokens/sec, TTFT, peak memory, energy per 1000 tokens, accuracy parity check.
- **Experiments:** convert top configs to GGUF/MLX, benchmark on the MacBook, compare against the GPU numbers. Expect MLX to lead on Apple Silicon for dense sub-14B models (independent benchmarks report MLX 20-87% faster than llama.cpp under 14B).
- **"Good result" even if hypothesis wrong:** a cross-platform reproducibility report is a strong, rare artifact for a CV.

Methodology

Candidate base models (all sub-8B, open-weight, as of 2026)

- **Qwen2.5 (0.5B, 1.5B, 7B):** primary spine. Per the official Qwen2.5 release blog, "All our open-source models, except for the 3B and 72B variants, are licensed under Apache 2.0." So the 0.5B/1.5B/7B are clean Apache 2.0; AVOID the 3B (Qwen

research licence) for the cleanest legal story. (Qwen)

- **Llama 3.2 (1B, 3B):** Llama Community Licence. The MAU cap is specifically 700 million: if you exceed "700 million monthly active users in the preceding calendar month, you must request a license from Meta." Also requires "Built with Llama" attribution and carries EU restrictions on some variants. Cross-family check only.

(MySQL)

- **Gemma 2 (2B):** Gemma Terms of Use (commercial use permitted, Google reserves remote restriction rights). Cross-family check.
- **Phi-3.5-mini (3.8B):** MIT licence, strong reasoning/math for its size.
- **SmolLM2 (1.7B):** Apache 2.0, fast iteration.

Recommendation: fix Qwen2.5-0.5B/1.5B/7B as the core spine (clean Apache 2.0, spans a wide size range for size-effect analysis), then add one Llama and one Phi model for cross-family generalisation in RQ2/RQ3.

Datasets and benchmarks

- **Reasoning/knowledge:** MMLU (or a fixed MMLU subset for speed), ARC-Challenge, HellaSwag, GSM8K (math), BBH.
- **Instruction-following:** IFEval (a known quantisation-sensitive task, so central to your story).
- **Domain-specific (pick one):** a code set (HumanEval/MBPP) or a biomedical/legal QA subset for a domain angle.
- **Calibration/robustness:** compute ECE from benchmark logits; for robustness apply input perturbations (typos, paraphrase) to a subset.
- **Calibration data for PTQ:** WikiText-2 and C4 (standard 128-sample calibration sets). Document the calibration source explicitly: prior work shows GPTQ specifically is sensitive to calibration-set domain, while AWQ is more robust to it.

Compression/efficiency techniques to compare

- **Quantisation:** GPTQ (AutoGPTQ/llm-compressor), AWQ (autoawq), bitsandbytes NF4 (4-bit); optionally SmoothQuant (W8A8). AWQ generally beats GPTQ on 4-bit weight-only accuracy; GPTQ is calibration-sensitive.
- **Pruning:** Wanda (single forward pass, memory-light) and SparseGPT, at 50% unstructured and 2:4 semi-structured. Flag that 50% unstructured sparsity gives essentially no GPU speedup without sparse kernels; 2:4 is what enables Ampere/Hopper sparse tensor cores.
- **PEFT:** LoRA, QLoRA (4-bit base + LoRA), DoRA, via the PEFT library.
- **Distillation (scoped):** sequence-level / response-based KD from a 7B teacher to a smaller student on one task dataset. Keep KD lightweight; it is the most compute-hungry component.

Evaluation metrics

- **Accuracy:** task-specific (exact_match, acc, acc_norm, pass@k) via lm-evaluation-

harness (reports acc_stderr automatically).

- **Calibration:** ECE, confidence on true labels (motivated by Proskurina et al., "When Quantization Affects Confidence of Large Language Models?", Findings of NAACL 2024 / arXiv:2405.00632, which found GPTQ 4-bit "results in a decrease in confidence regarding true labels" and disproportionately affects already-low-confidence samples). ([arxiv](#))
- **Efficiency:** FLOPs/MACs per token (DeepSpeed FLOPs profiler or PyTorch FlopCounterMode; note ptflops/fvcore do NOT handle autoregressive decode with KV-cache), tokens/sec, TTFT, peak memory (torch.cuda.max_memory_allocated), disk size. Report wall-clock alongside FLOPs, since decode is memory-bound and FLOPs alone mispredict speed.
- **Energy:** CodeCarbon or Zeus on the GPU (both read NVML/nvidia-smi). On Apple Silicon use powermetrics or report wall-clock and flag the limitation.

Tools and what works where

- **Works on CUDA (Colab):** transformers, peft, AutoGPTQ, autoawq, bitsandbytes, SparseGPT/Wanda repos, vLLM, lm-evaluation-harness, DeepSpeed FLOPs profiler, CodeCarbon, Zeus.
- **Works on Apple Silicon MPS:** transformers (MPS backend), MLX/mlx-lm (Apple's native framework, fastest local inference on Apple Silicon for dense sub-14B models), llama.cpp/GGUF (Metal backend), lm-evaluation-harness against a local GGUF/MLX server.
- **Does NOT work on MPS:** bitsandbytes (CUDA-only), AutoGPTQ/autoawq kernels (CUDA), vLLM (CUDA). So do all quantised-training/PTQ and CUDA-kernel work on Colab; do MLX/GGUF conversion and on-device latency/energy on the Mac.

Feasibility and compute plan

- **Colab reality:** free tier gives a T4 (~15GB) but disconnects after ~90 min idle and caps at ~12h/session with no guaranteed GPU. Colab Pro is \$9.99/month for exactly 100 compute units; a T4 burns roughly 1.96 units/hour (about 51 T4-hours/month), and even paying users can be handed a T4 rather than an L4/A100. Plan for interruptions: checkpoint everything to Google Drive, keep runs under ~2-3h, script for resumability. ([Thundercompute + 4](#))
- **What fits:** PTQ (GPTQ/AWQ/NF4) on 0.5B-7B fits a T4. QLoRA of a 7B fits a single consumer GPU. Wanda/SparseGPT prune layer-by-layer so 7B fits in modest memory. Evaluation is the time sink, not training.
- **What to watch:** full 57-subject MMLU is slow; use a fixed subset or fewer few-shot examples. KD is the most expensive; scope to one teacher/student pair on one dataset.
- **Fallbacks:** if a run is too big, drop to 0.5B/1.5B; cut MMLU to a subset; reduce seeds from 5 to 3; drop SmoothQuant; make KD optional (RQ3 can stand on quant x prune x LoRA interactions alone).

- **Approximate experiment counts:** RQ1 ~6 baselines. RQ2 ~6 models x ~6 methods. RQ3 factorial ~3 method pairs x 2 orderings x 3 models x 3 seeds. RQ4 top ~5 configs x 3 runtimes. Order of a few hundred evaluation runs; batch and script them.
- **Runtime estimate:** PTQ minutes to tens of minutes per model; QLoRA a few hours per run; each full eval tens of minutes. Budget ~2-3 focused compute days per RQ spread across the term.

Term-by-term work plan

Advanced Computer Science Masters Project (29 Sep 2026 - 16 Jan 2027): scoping, literature, pipeline, RQ1+RQ2

- Weeks 1-3: finalise title/RQs with supervisor; write the literature review; lock model/benchmark/tool list.
- Weeks 4-6: build the reproducible pipeline (config management, seeds, logging, Drive checkpointing); reproduce one published baseline number to validate the harness.
- Weeks 7-11: run RQ1 baselines and RQ2 single-method experiments; build Pareto plots.
- Weeks 12-15: write Methods + RQ1/RQ2 chapters. Milestone: working pipeline + baseline tables + method comparison.
- Risk cut: if behind, drop cross-family models here and keep the Qwen spine.

Advanced Research Topics in Computer Science (20 Jan 2027 - 31 May 2027): RQ3 (core) + start RQ4

- Weeks 1-6: run the RQ3 factorial (stacking/ordering) with seeds; compute interaction matrices and significance.
- Weeks 7-10: analyse synergy/antagonism; write the RQ3 chapter (your strongest novelty).
- Weeks 11-15: begin RQ4: convert top configs to GGUF/MLX; first Apple Silicon benchmarks. Milestone: interaction results + first on-device numbers.
- Risk cut: if behind, reduce to two method pairs and one ordering contrast.

Research Methods (25 May 2027 - 30 Sep 2027): finish RQ4, statistics, write-up, release

- Weeks 1-4: finish RQ4 on-device latency/energy; cross-platform comparison.
- Weeks 5-8: statistical analysis (bootstrap CIs, multi-seed variance), ablations, limitations.
- Weeks 9-14: full dissertation write-up; polish figures; finalise GitHub artifact and model cards.
- Weeks 15+: buffer for proofreading and submission. Milestone: submitted dissertation + released reproducible benchmark.

How to do the research properly at UK MSc level

- **Literature strategy:** search arXiv (cs.CL, cs.LG), ACL Anthology, OpenReview (ICLR/NeurIPS), MLSys, and the NeurIPS ENLSP (Efficient Natural Language and Speech Processing) workshop. Anchor on recent surveys plus the two gap papers (Zhou/Kurz/Zhao EMNLP 2025 on SLM compression dynamics; Chhawri et al. arXiv:2511.19495 on compression ordering). Forward-cite chase via Google Scholar. Track everything in a reference manager.
- **Research design:** state hypotheses; hold compression ratio constant when comparing methods (fair comparison); include controls (full-precision baseline) and ablations (each method alone before stacking); fix and report seeds; run multiple seeds for anything with training.
- **Statistics for small-n ML:** use non-parametric bootstrap confidence intervals over the eval set (10k resamples; do not use bootstrap with fewer than ~20 samples); report the harness's built-in stderr; for seed variance use 3-5 seeds and report mean plus/minus CI. Be honest that a +1% difference with 3 seeds is often not statistically significant. Do not over-claim. (arxiv) (arxiv)
- **Reproducibility:** pin environments (requirements.txt / Docker), fix seeds, version datasets, save all configs, write model cards, release code and result JSONs. Report exact library versions, since quantisation results shift across releases.
- **Ethics/legal:** document every model and dataset licence (Qwen2.5 Apache 2.0 except 3B/72B; Llama Community Licence with the 700M MAU cap and attribution/EU terms; Gemma Terms of Use); report energy/carbon via CodeCarbon; note "open weight" is not OSI "open source."
- **Dissertation chapters:** Introduction; Literature Review; Methodology; Experimental Setup; Results (one section per RQ); Discussion; Conclusion and Future Work; Appendices (code, configs, full tables).
- **Why UK MSc CS dissertations lose marks:** description instead of critical analysis; weak/unjustified methodology; thin evaluation rigour; poor time management leading to a rushed write-up; shallow literature engagement. UK CS examiners explicitly reward evaluation quality (choice of methods, controls, conditions, rigour, analysis of data) and engagement with recent literature; novelty is not required but rigour is.

CV / employability mapping

- **CV bullet points this produces:**
 - "Built a reproducible benchmark measuring accuracy, calibration, FLOPs, latency, memory, and energy trade-offs of stacked LLM compression (quantisation, pruning, LoRA/QLoRA/DoRA, distillation) across sub-8B open-weight models."
 - "Quantified interaction effects and optimal ordering of stacked compression methods on consumer GPU and Apple Silicon; released as open-source code and model cards."

- "Profiled accelerator inference (tokens/sec, TTFT, peak memory, energy) and cross-platform deployment (CUDA transformers/vLLM vs MLX vs llama.cpp Metal)."
- **Skill keywords:** LLM inference optimisation, model compression, quantisation (GPTQ/AWQ/NF4), pruning (Wanda/SparseGPT), PEFT (LoRA/QLoRA/DoRA), knowledge distillation, GPU/accelerator profiling, FLOPs/MACs, calibration/ECE, lm-evaluation-harness, vLLM, MLX, llama.cpp/GGUF, reproducible ML, MLOps.
- **Maps to UK entry-level ML/AI job requirements:** LLM optimisation and serving, benchmarking/experimentation, PyTorch, deployment on constrained hardware, evaluation rigour. UK listings repeatedly ask candidates to "conduct experiments and benchmarking to assess the performance of various model architectures and optimize hyperparameters" and to "optimize existing LLM training and serving."
- **Packaging:** clean GitHub repo (README, reproducible configs, one-command eval, results JSON, plots, model cards, licence file); a short technical blog post or preprint; a downloadable benchmark artifact.
- **Publishability:** the RQ3 interaction/ordering results are realistically a workshop paper (NeurIPS ENLSP or a similar efficiency workshop, 4-page format), especially with calibration/robustness outcomes that most compression papers omit.

Alternative adjacent backup topics

- **Calibration-focused variant:** "How compression reshapes uncertainty in small LMs." Narrower, novel because most compression work ignores calibration, and cheap to run.
- **On-device runtime shoot-out:** "MLX vs llama.cpp vs Core ML for sub-4B models on Apple Silicon." Strong systems/deployment angle, minimal training compute, very CV-friendly.
- **Calibration-data sensitivity of PTQ:** "Does calibration-set domain change GPTQ/AWQ outcomes for small models?" Tight, controlled study exploiting a known open question (GPTQ is calibration-sensitive, AWQ less so), easily single-student-sized.

Caveats

- Some 2026 sources are recent preprints (e.g. arXiv:2511.19495); verify final published numbers before citing.
- Robustness findings are mixed: some studies find quantisation does NOT disproportionately hurt hard tasks and can even help robustness to natural perturbations. Jin et al. (arXiv:2409.11055) report quantized models "maintain similar performance to full-precision models across both easy and challenging tasks," while half-precision models did better on TruthfulQA and IFEval. Do not assume compression uniformly hurts. (arXiv)
- Colab availability is genuinely unreliable (90-min idle disconnect, ~12h cap, no

guaranteed GPU even on Pro); the plan must assume interruptions and checkpoint aggressively.

- Energy measurement on Apple Silicon is approximate; report method and limitations rather than precise joule claims. CodeCarbon estimates on GPU can also deviate from ground truth due to cooling/peripheral overheads.
- Model licences change over generations (Gemma moved to Apache 2.0 in a later release; Qwen's newest models pulled back from full openness); re-check terms at the time of use.